

Evaluation of the IMT-2020 candidate radio interface technology DECT-2020 NR

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Abstract—This paper provides an overview of DECT-2020 NR technology evaluation conducted by the Wireless World Research Forum (WWRF) in the context of IMT-2020 Evaluation Process solicited by the International Telecommunication Union (ITU). DECT-2020 NR, a Radio Interface Technology developed by ETSI TC DECT and DECT Forum, was submitted as an IMT-2020 candidate along with five other technologies. It is applicable in various use cases and markets, with focus on consumer and industrial IoT applications, such as industry and building automation and monitoring. In particular, it is targeting Ultra-Reliable Low Latency (URLLC) and massive Machine Type Communication (mMTC) scenarios, based on decentralized network deployments. The WWRF performed a complete assessment of the technology which included a high-level assessment of services and spectrum capabilities, along with a minimum technical performance requirements analysis for the URLLC and mMTC test environments, based on inspection, analysis and simulation. For assessing characteristics using simulation, i.e., reliability and connection density, an in-house simulation tool was developed. The evaluation study addressed mesh topologies supported by DECT-2020 and proposed several enhancements to the ITU's evaluation methodology, which is defined for cellular topologies only. The technology was found to fulfill all the requirements specified by ITU's evaluation process for IMT-2020.

Index Terms—ITU evaluation, DECT-2020 NR, IMT-2020, URLLC, mMTC

I. INTRODUCTION

Working Party 5D of the International Telecommunication Union (ITU WP 5D) is responsible for the specification of the overall radio system aspects of the terrestrial component of International Mobile Telecommunications (IMT) systems, currently comprising of IMT-2000, IMT-Advanced, and IMT-2020 - commonly known outside ITU as 3G, 4G and 5G respectively. ITU recognizes a technology as IMT-2020 (and consequently 5G), if it is a mobile system that proposes one or more new radio interfaces, able to support new capabilities beyond IMT-advanced. Therefore, the ITU WP 5D has defined an 8-step process from technology preparation and development to assessment, leading to an ITU recommendation acknowledging or not, a proposal as an IMT-2020 technology.

As part of Step 3, that includes submission of new Radio Interface Technologies (RITs), six candidate RITs have been

submitted and qualified for further evaluation by WP 5D in 2019. Step 4 of the process, includes elaborate evaluation of the proposed RITs by Independent Evaluation Groups (IEGs). After detailed examination of the technologies, the IEGs should conclude, if the IMT-2020 requirements are covered or not. The evaluation studies are performed on the basis of a methodology formulated by WP 5D and potentially additional methodologies proposed by the various IEGs. The evaluation is based on the characteristics defined in ITU-R Reports M.2410-0 [1], M.2411-0 [2] and M.2412-0 [3] using a methodology described in Report ITU-R M.2412-0. The methodology aims at assessing the service, spectrum and technical characteristics for the candidate RITs, given a set of minimum performance requirements.

During 2020, it was concluded that the set of RITs (SRIT) proposed by 3GPP (Release 16 [4]) should proceed to next steps of the process. Along with 3GPP, some variants relying on it, submitted by Republic of Korea, People's Republic of China and India (TSDSI) were also qualified. However, at the same time, the proponent's ETSI (TC DECT) & DECT FORUM submission for the DECT-2020 NR (Digital Enhanced Cordless Telecommunications) radio interface did not receive a complete evaluation as to whether it satisfies all the requirements for a RIT component. As a way forward, WP 5D decided to extend the evaluation process for the ETSI (TC DECT) & DECT FORUM. Thus, at the end of 2020, an open request to IEGs for the evaluation of the remaining SRITs were issued with the objective to finalize the procedure by October 2021.

The Wireless World Research Forum (WWRF) was one of the IEGs that stepped up for the evaluation process. Over the last years, WWRF championed several activities focused on the wireless evolution to and beyond 5G. On December 2020, WWRF has been accepted as a Sector Member of ITU. In its capacity as ITU member WWRF had many direct contributions, including its participation as an IEG in the IMT-2020 Step 4 evaluation of the TSDSI and EUHT (NUFRONT) technology proposals, during 2019-2020. WWRF contributed three reports -related to DECT-2020 NR technology - leading to the final report on August 2021. In this paper, the results and conclusion of these studies for the DECT NR-2020 RIT

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are presented. A short introduction to the RIT is presented in Section II. In Section III, the ITU requirements and the work methodology are presented, while in Section IV the analysis results are provided. Finally, conclusions are derived in Section V.

II. OVERVIEW OF THE DECT-2020 NR TECHNOLOGY

DECT-2020 NR has been specified in the following series of radio communication standards:

- TS103.636-1 [5], a document that provides an overview of the standard;
- TS103.636-2 [6], that establishes the minimum RF requirements for DECT-2020 NR Radio Devices;
- TS103.636-3 [7], that specifies the transmission techniques in the DECT-2020 NR physical layer;
- TS103.636-4 [8], that specifies the medium access procedures.

DECT-2020 NR is a non-cellular RIT applicable in various use cases and markets. It mainly targets consumer and industrial IoT supporting and enabling industry and infrastructure automation and monitoring. It is positioned as an Ultra-Reliable Low Latency (URLLC) and massive Machine Type Communication (mMTC) enabler as defined by ITU-R for IMT-2020. The technology supports autonomous operation, while if necessary, interworking with Wide Area Networks (WAN - e.g., PLMN, fibre, satellite, etc.) is possible.

According to [5], DECT-2020 NR can be used as a foundation for:

- Very reliable Point-to-Point and Point-to-Multipoint Wireless Links;
- Local Area Wireless Access Networks in a typical DECT star topology supporting URLLC scenarios;
- Mesh and Self-Organizing Local Area Wireless Access Networks following a mesh network topology, enabling mMTC use cases.

The DECT-2020 NR physical layer [7] is in principle designed for sub-6GHz frequency bands. The physical layer employs Orthogonal Frequency Division Multiplexing (OFDM) modulation with both Time and Frequency Division Multiple Access (TDMA and FDMA respectively) techniques supported. Duplexing is performed with time division (TDD). DECT OFDM supports multiple numerologies, allowing operation with different channel bandwidths from approximately 1.5MHz to 200MHz and subcarrier spacing varying from 27kHz to 216kHz, providing configurations for various propagation conditions. Standardized precoders for up to 8 streams MIMO operation are also supported (however, beamformers beyond the standard can be used, as long as they do not impose extra requirements at the receiver). DECT-2020 NR is designed to enable coexistence with legacy DECT and DECT evolution in frequency bands currently allocated to DECT.

DECT-2020 NR does not follow the conventional cellular structure. Radio devices (RDs) can operate in two modes:

- Fixed Termination points (FT), initiating coordination of local radio resources, and providing information on how other RDs may connect and communicate with it;

- Portable Termination points (PT), that select another RD, which is in FT mode, for association.

It is noted that apart from their scope, the standard does not distinguish physical layer functions between FTs and PTs.

One of the most distinctive features of DECT-2020 NR is the mesh networking capability [8], which allows for network topologies and deployments in IoT / mMTC usage scenarios, addressing the link budget limitations of classical cellular technologies. In particular, mesh network devices can communicate directly to each other extending the range of network and increasing the reliability of communication. The role of the involved RDs may change autonomously. Each RD can act as a node transmitting a message, as a node forwarding any message from another radio device or as a node being the destination of a message. Each RD can communicate directly (device to device) or, if not in range, indirectly - via other radio devices establishing a communication route - which minimizes the probability of outage.

Mesh operation supports autonomous routing using a cost value, without the need to maintain routing tables in each device. To achieve scalability the RDs are able to:

- route data;
- take local decisions of the radio resources;
- operate with no need for central coordination;
- change roles between routing and non-routing nodes.

The mesh system operation is based on a clustered tree topology. Each RD decides the next hop individually based on the currently available routes towards the RD providing the connectivity to the internet (FT). Each RD has knowledge of the next uplink and downlink hop, and each cluster controls radio resources independently. An example of the clustered topology is shown in Fig.1 [5].

The RD considers the route cost, the cost of the route to deliver data to the sink, i.e., to the FT that has a backend connection to the internet. RD may select the RD for association which indicates the smallest cost. The route cost calculation is left to implementation. Route cost value will increase at least 1 in every hop. The maximum route cost value is 254, which is not seen as a limiting factor in real deployments.

A. Submission

ETSI (TC DECT) & DECT Forum submitted an SRIT proposal that includes the following two component RITs:

- DECT-2020 NR RIT component
- 3GPP NR RIT component

The 3GPP NR component has already been evaluated by other IEGs. Hence, WWRF along with other IEGs decided to evaluate only the DECT-2020 NR RIT component. More specific, ITU-R imposes requirements for three usage scenarios: enhanced Mobile BroadBand (eMBB), URLLC, and mMTC. Within the SRIT, the eMBB usage scenario is addressed by the 3GPP NR component, while the DECT-2020 NR component address two usage scenarios (mMTC and URLLC) as described in [3].

Since the DECT-2020 component RIT addresses URLLC and mMTC test environments, only the technical performance

ETSI TS 103 636-1 V1.1.1 (2020-07)

TABLE I
TECHNICAL PERFORMANCE REQUIREMENTS

Characteristic for evaluation (test-environment)	High-level assessment method	Reference to [1]	Reference to [3]
User plane latency (URLLC)	Analytical	§ 4.7.1	§ 7.2.6
Control plane latency (URLLC)	Analytical	§ 4.7.2	§ 7.2.5
Mobility interruption time (URLLC)	Analytical	§ 4.12	§ 7.2.7
Reliability (URLLC)	Simulation	§ 4.10	§ 7.1.5
Connection density (mMTC)	Simulation	§ 4.8	§ 7.1.3
Bandwidth	Inspection	§ 4.13	§ 7.3.1

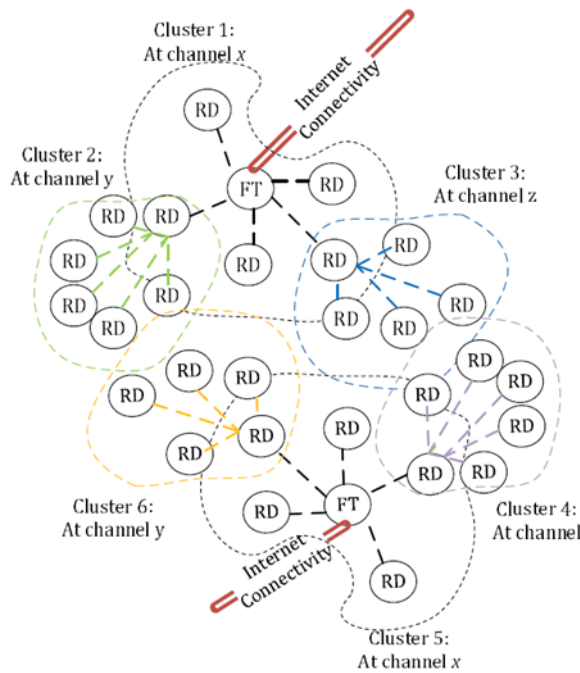


Fig. 1. The DECT-2020 NR mesh clustered tree topology

requirements related to these test environments have been evaluated.

III. REQUIREMENTS AND METHODOLOGY

A. Requirements

The evaluation methodology includes two core parts:

- A high-level assessment of services and spectrum capabilities;
- A minimum technical performance requirements analysis.

Regarding the latter, the technical performance requirements listed in the Table I are assessed. In this Table, for each addressed characteristic the assessment method used (based on ITU evaluation methodology) is listed with references to the requirements and exact methodology steps extracted from corresponding ITU Documents [1] and [3].

As a summary, some basic requirement values are provided (the reader is encouraged to study the details in [1] and [3]):

- **User-plane latency** should be 1ms for both downlink and uplink (unloaded conditions) for URLLC.
- **Control-plane latency** should be lower than 20ms.
- **Mobility interruption time** should be 0 ms.
- **Reliability** requirement is 10^{-5} probability of failure in transmitting a layer 2 Protocol Data Unit (PDU) of 32 bytes within 1 ms in channel quality of coverage edge for the Urban Macro-URLLC test environment.
- **Connection density** requirement is fulfilled when 1,000,000 devices per km^2 are able to deliver a message

of certain size in a given time window. The time window is defined as 24 hours (loose requirement) or 2 hours (strict requirement).

- **Bandwidth** requirement is at least 100 MHz, with the possibility to achieve 1 GHz overall with multiple carriers.

B. Method of work

For assessing characteristics using simulation, i.e., reliability and connection density, we rely on an in-house simulation tool. An End-to-End Physical Layer simulator of the DECT-2020 NR technology was developed by WWRF IEG in MATLAB/Octave, able to perform jointly system-level and link-level experiments. The developed simulator is decomposed into two main components:

- The link level simulator: implementing the radio transmission between two nodes of the system.
- The system level simulator: implementing the radio access network setup according to the specifications of the ITU evaluation guidelines ([3] § 8.4)

It is noted that both simulator components are developed in MATLAB/Octave. The system simulation procedure for the evaluation and validation of the KPI for connection density is detailed in [3], for cellular topology only. By relying on the cellular topology DECT-2020 NR will fail to meet the requirements. However, as mentioned before, the DECT-2020 NR RIT introduces a mesh network topology and system operation extending the range of network and increasing the reliability of communication.

In Fig.2, the system level simulator initialization and utilization process is presented, focusing on the URLLC reliability use case, i.e., the system level simulator is configured to fit the specifications provided by ITU for the URLLC setup.

Based on the ITU specifications and guidelines for URLLC reliability:

- A star topology with coordinated, scheduled transmission is considered.
- A 19-cell scenario is considered – with a three sectoral base stations setup. The PTs are considered directly connected with the FTs.

TABLE II
KPIs RELATED TO URLLC AND mMTC USAGE SCENARIOS EVALUATED FOR THE DECT-2020 NR COMPONENT

Minimum technical performance requirements	Category			Required value	Value	Requirement met
	Usage scenario	Test environment	Downlink or uplink			
User plane latency (ms)	URLLC	Not applicable	Uplink and Downlink	1	0.198-0.975 ms	Yes
Control plane latency (ms)	URLLC	Not applicable	Not applicable	20	6.25ms	Yes
Connection density (devices/km ²)	mMTC	Urban Macro – mMTC	Uplink	1 000 000 users with less than 1% outage for a) one message per day, b) one message per 2 hours	More than 1M users with less than 1% outage with message transmission period of less than 10 minutes	Yes
Reliability	URLLC	Urban Macro –URLLC	Downlink	$1 - 10^{-5}$ success probability of transmitting a layer 2 PDU (protocol data unit) of size 32 bytes within 1 ms in channel quality of coverage edge	$>1 - 10^{-6}$ for both LoS and NLoS for Configuration B with 5x4 with 1/7 frequency reuse and 1.728MHz bandwidth	Yes
Mobility interruption time (ms)	eMBB and URLLC	Not applicable	Not applicable	0	0 ms	Yes
Bandwidth and Scalability	All	Not applicable	Not applicable	At least 100 MHz Up to 1 GHz Support of multiple different bandwidth values	221,184 MHz Multiple Carriers 1.728 MHz to 221.184 MHz.	Yes Yes Yes

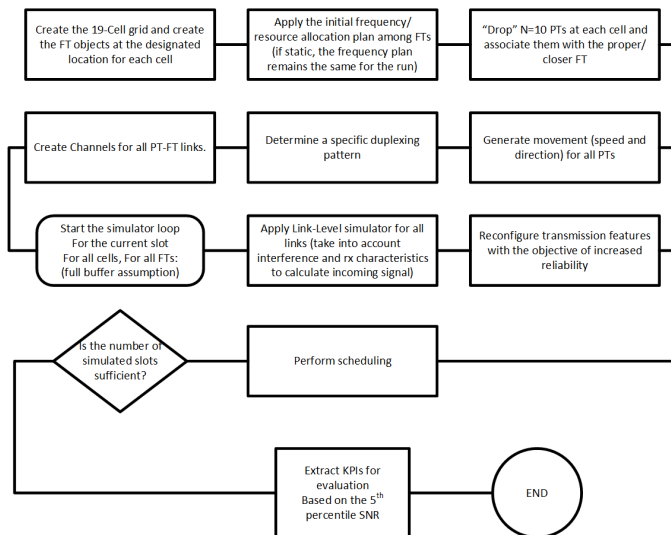


Fig. 2. System Level Simulator functional flow

- Four channel allocation procedures are considered (1-channel, 3-channel and 7-channel reuse plan).
- Scheduled channel allocation and/or cognitive channel selection using measurement is considered.

Since, no throughput requirement is specified in the URLLC scenario, time slot sharing among cells is also allowed, i.e., cells with the same carrier frequency may use different times-

lots in order to further improve interference isolation

Since [3] does not address the methodological framework for mesh topologies, WWRF IEG devoted significant resources for rigorously defining all the additional aspects required for performing such an evaluation. As a result, a configuration and a simulation procedure are determined that is able to put the DECT-2020 NR RIT and its adhoc/relaying component under test according to the principles and objectives of the M.2412 guidelines.

WWRF IEG addressed the following issues:

- System model definition/extension including relay/forwarding nodes in addition to standard cellular UEs.
- Introduction of a simple but effective protocol for multi-hop communications.
- Definition of a relay selection scheme based on channel quality conditions.
- Consideration of device-to-device channel models for relay communication, not previously considered in cellular technologies evaluation

WWRF IEG developed a system model simulator that is able to take into account the particularities of the DECT-2020 NR RIT, while remaining aligned with the evaluation guidelines defined in ITU-R Report [3].

TABLE III
SIMULATED CASES AND SCENARIOS FOR THE EVALUATION OF THE
RELIABILITY KPIS FOR THE DECT-2020 NR COMPONENT

Setup	5th percentile point packet error probability	SINR threshold	Fulfilment of the requirement
SISO with 1/7 frequency reuse and 1.728MHz bandwidth (Configuration B)	approximately 0.05 for LoS; approximately 0.5 for NLoS	3.2 dB	NO
2x2 with 1/7 frequency reuse and 1.728MHz bandwidth (Configuration B)	$<10^{-6}$ for LoS; approximately 0.05 for NLoS	3.2 dB	YES (for LoS)
5x4 with 1/7 frequency reuse and 1.728MHz bandwidth (Configuration B)	$<10^{-6}$ for LoS (10^{-5} is achieved at appr. -8 dB); $<10^{-6}$ for NLoS	3.2 dB	YES
SISO with 1/3 frequency reuse and 1.728MHz bandwidth (Configuration B)	Nonfunctional at this SNR	-5.1 dB	NO
2x2 with 1/3 frequency reuse and 1.728MHz bandwidth (Configuration B)	approximately 0.02 for LoS; Nonfunctional at this SNR for NLoS	-5.1 dB	NO
5x4 with 1/3 frequency reuse and 1.728MHz bandwidth (Configuration B)	$<10^{-6}$ for LoS (10^{-5} is achieved at appr. -8 dB); approximately 0.005 for NLoS	-5.1 dB	YES (for LoS)

IV. RESULTS

Based on the aforementioned method of work, all KPIS related to URLLC and mMTC usage scenarios are evaluated for the DECT 2020 NR component. The summary of results of the evaluation process are presented in Table II. Details of the WWRF evaluation report can be found in [9].

The results, through inspection, analysis and simulation, verify that DECT-2020 NR RIT fulfills the IMT-2020 requirements for URLLC and mMTC, and therefore, since the SRIT also contains 3GPP 5G NR for eMBB use cases, it qualifies for the next steps of the ITU-R process.

If we focus on the URLLC reliability use cases, extensive simulations were performed for various configurations. In some cases the requirements were met, while in others the system was not able to cover them. Various antenna schemes were considered for the FTs in order to enable beamforming capabilities. It is noted that only through multi-antenna operation, the RIT is able to cover the reliability requirements. Results are presented in Table III. In the selected set of results, the FTs are considered to be equipped with a single antenna, a 2x2 rectangular panel antenna (4 elements) and a 5x4 rectangular panel antenna (20 elements). Configuration B refers to the second system level configuration defined by ITU-R in [3].

V. CONCLUSIONS

WWRF IEG performed a complete evaluation of the DECT-2020 NR RIT component as part of the re-engagement in Step 4 ITU-R evaluation. The main conclusions were:

- 1) The proposed RIT component is part of an SRIT which also includes the 3GPP 5G NR [4], which was concluded by WP 5D to fulfill the requirements. Therefore, for the RIT evaluation only URLLC and mMTC test environments were considered.
- 2) Assessment was based on the RIT submission documents and published technology specifications by ETSI [5] - [8], following the ITU-R guidelines [1], [2], [3].
- 3) The DECT NR 2020 RIT supports URLLC and mMTC usage scenarios.
- 4) The DECT NR 2020 RIT fulfils the requirements and parameters that are evaluated by inspection, i.e. bandwidth, spectrum, and services.
- 5) The DECT NR 2020 RIT fulfils the requirements that are evaluated by analysis, i.e. user-plane latency, control-plane latency and mobility interruption time.
- 6) The DECT-2020 NR RIT is able to satisfy the reliability requirements for various configurations, as demonstrated through simulation.
- 7) The DECT-2020 NR RIT is able to satisfy the connection density requirements for various configurations, as demonstrated through simulation with the use of multi-hop communications and a properly defined relay protocol.

Based on the evaluation, the DECT-2020 NR RIT was found to fulfil all the requirements specified by ITU's evaluation process for IMT-2020.

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